

BENSON, United Kingdom

WMO: 036580

Lat: 51.6203N

Lon: 1.0986W

Elev: 57

StdP: 100.64

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.5	(c) -3.7	(d) -7.2	(e) 2.1	(f) -3.8	(g) -5.5	(h) 2.4	(i) -2.0	(j) 12.9	(k) 9.0	(l) 11.7	(m) 9.6	(n) 1.1	(o) 50	(p) 0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.7	(c) 27.5	(d) 18.3	(e) 25.2	(f) 17.5	(g) 23.4	(h) 16.7	(i) 19.4	(j) 25.1	(k) 18.4	(l) 23.2	(m) 17.6	(n) 21.8	(o) 4.1	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.4	12.5	21.1	16.6	11.9	20.1	15.8	11.3	19.3	55.3	24.9	52.1	23.3	49.5	21.8	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	9.3	8.3	-9.3	30.6	2.1	2.2	-10.8	32.2	-12.1	33.4	-13.2	34.7	-14.8	36.3
			-9.7	21.2	2.1	1.1	-11.2	22.0	-12.4	22.6	-13.6	23.2	-15.1	24.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	4.3	4.6	6.3	8.6	11.6	14.7	17.0	16.7	14.0	10.9	7.1	4.9
	DBStd	5.53	3.62	3.45	3.12	3.08	3.04	2.68	2.60	2.62	2.65	3.23	3.56	4.09
	HDD10.0	818	176	152	118	61	18	1	0	0	2	28	99	163
	HDD18.3	3046	433	383	372	291	208	114	58	66	132	232	338	417
	CDD10.0	851	1	1	5	19	68	141	218	206	122	54	11	4
	CDD18.3	38	0	0	0	0	1	4	17	14	2	0	0	0
	CDH23.3	429	0	0	0	2	9	64	203	131	18	2	0	0
	CDH26.7	91	0	0	0	0	0	12	47	31	1	0	0	0
Wind	WSAvg	3.8	4.3	4.3	4.1	3.8	3.8	3.5	3.4	3.3	3.5	3.9	3.7	4.1
Precipitation	PrecAvg	675	66	49	41	53	53	49	49	57	54	68	74	63
	PrecMax	878	154	115	88	138	132	136	112	130	111	136	144	104
	PrecMin	493	12	6	11	3	6	9	11	6	10	24	27	13
	PrecStd	116	32	28	21	34	27	28	25	33	30	32	32	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	13.7	16.9	22.0	24.5	28.4	30.4	29.8	25.7	20.7	15.7	14.1
		MCWB	11.4	11.1	11.2	14.1	16.5	18.8	19.0	19.3	17.7	16.6	13.9	12.3
	2%	DB	12.0	12.0	14.5	18.5	22.1	25.1	27.8	26.9	22.9	18.1	14.5	12.9
		MCWB	10.5	9.9	10.3	12.6	15.9	17.4	18.2	18.5	16.8	14.7	12.8	11.6
	5%	DB	10.9	10.9	12.9	16.2	19.9	22.7	25.5	24.3	20.8	16.9	13.5	12.0
		MCWB	9.4	9.0	9.7	11.3	14.7	16.4	17.5	17.3	15.6	14.2	11.9	10.7
	10%	DB	9.8	9.9	11.6	14.3	17.6	20.7	23.4	22.2	19.2	15.8	12.4	11.0
		MCWB	8.5	8.2	9.0	10.3	13.2	15.4	16.8	16.4	15.0	13.6	10.9	9.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	11.8	12.2	15.3	18.3	20.1	20.4	20.7	19.1	17.0	14.4	12.7
		MCDB	12.6	13.4	14.7	20.4	22.4	26.7	27.4	27.4	23.3	19.9	15.4	13.4
	2%	WB	10.8	10.3	11.1	13.2	16.5	18.2	19.2	19.2	17.6	15.7	13.2	11.9
		MCDB	11.9	11.6	13.4	17.4	20.8	22.9	25.3	24.4	21.3	17.4	14.2	12.8
	5%	WB	9.6	9.4	10.4	11.9	15.1	17.1	18.3	18.2	16.7	14.8	12.1	10.9
		MCDB	10.7	10.7	12.4	15.2	19.0	21.4	23.7	22.3	19.6	16.3	13.2	12.0
	10%	WB	8.7	8.5	9.4	10.9	13.9	16.1	17.4	17.2	15.7	14.0	11.1	9.8
		MCDB	9.7	9.8	11.4	13.7	17.2	20.1	22.2	20.9	18.3	15.5	12.3	10.9
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.8	8.4	9.8	9.8	10.2	10.7	10.2	9.7	8.1	6.8	6.1
		MCDBR	6.5	7.0	10.2	12.8	13.4	14.0	14.8	13.6	12.1	8.3	6.2	5.5
		MCWBR	6.0	5.7	6.8	7.7	7.8	7.5	6.5	6.2	6.7	5.7	5.2	5.2
	5% WB	MCDBR	6.4	6.7	8.9	11.5	11.6	12.1	12.6	11.2	10.1	7.3	6.1	5.5
		MCWBR	6.1	5.8	6.4	7.2	7.2	7.0	6.3	5.9	6.4	5.6	5.5	5.4
Clear-Sky Solar Irradiance	taub	0.326	0.342	0.378	0.401	0.405	0.403	0.399	0.389	0.372	0.355	0.329	0.316	
	taud	2.359	2.369	2.265	2.215	2.242	2.275	2.296	2.344	2.389	2.417	2.406	2.344	
	Ebn at Noon	676	774	808	833	848	852	849	838	810	748	669	620	
	Edh at Noon	63	81	108	127	130	127	122	111	95	77	60	55	
All-Sky Solar Radiation	RadAvg	0.81	1.47	2.54	3.91	4.75	5.24	4.97	4.13	3.11	1.81	0.97	0.64	
	RadStd	0.07	0.19	0.34	0.46	0.30	0.54	0.50	0.33	0.27	0.16	0.11	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (85 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page